

Cloud Smoke Test Results — 20 Epoch Run (Seed 1)

Date: 2026-03-15 -> 2026-03-16
Launched: ~21:20 PDT (March 15) via `./cloud/smoke_test.sh --epochs 20`
Purpose: Verify all training scripts run correctly on cloud VMs before full 5-seed production run.

SRNN Model Configuration

All experiments used identical SRNN hyperparameters:

Parameter	Value
Hidden size (n)	32
Excitatory neurons (n_E)	16
Inhibitory neurons (n_I)	16
SFA timescales (n_a_E)	3
STD timescales (n_b_E)	0 (disabled)
ODE solver	semi_implicit (fused)
ODE step size (h)	0.02 (1/50)
ODE unfolds per input step	6
Readout	synaptic
Per-neuron dynamics	false (shared scalars)
Total parameters	~19,180 (HAR; varies by input dim)
State dimension	80 (32 neurons × (1 V + 1 c + 3 a_E - 1 extra))

Training Settings

Parameter	Value
Optimizer	Adam
Learning rate	0.005
LR schedule	linear taper (final 20% of epochs)
Batch size	128
Epochs (smoke test)	20
Seed	1
Checkpointing	Best (by valid metric) + every 5 epochs

Cloud Infrastructure

Setting	Value
GCP Project	liquidneuralnets
Default VM type	n4d-highmem-2 (2 vCPU, 16 GB RAM)
Large VM type	n4d-highmem-4 (4 vCPU, 32 GB RAM)
VM image	srnn-julia-v1 (Julia 1.12.5, 545 packages precompiled)
Zone	us-central1-a
Spot VMs	No (standard — JIT too fragile for preemption)
Results bucket	gs://liquidneuralnets-experiments/results/

Run Status Summary

#	Experiment	Status	VM Type	Duration	RAM (RSS)
1	HAR	[DONE] Done (20/20)	n4d-highmem-2	305 min (5.1 hr)	~1.8 GB
2	Gesture	[DONE] Done (20/20)	n4d-highmem-2	38 min	—
3	Occupancy	[DONE] Done (20/20)	n4d-highmem-2	335 min (5.6 hr)	—
4	Ozone	[DONE] Done (20/20)	n4d-highmem-2	43 min	—
5	Traffic	[RUNNING] Running (12/20)	n4d-highmem-2	~50 min/epoch	9.6 GB
6	SMnist	[RUNNING] Running (2/20)	n4d-highmem-4	~3.5 hr/epoch	7.7 GB
7	Power	[RUNNING] Running (2/20)	n4d-highmem-4	~4.5 hr/epoch	8.8 GB
8	Person	[CRASHED] OOM (exit 137)	n4d-highmem-2	crashed <1 epoch	>16 GB
9	Cheetah	[FAILED] Not launched	—	—	—

Note: Only 8 of 9 experiments launched due to an external IPv4 address quota of 8. Person crashed with OOM on 16 GB VM; needs n4d-highmem-4 (32 GB). Env file updated for next run.

Results vs. Hasani et al. 2021 Table 3

Table 3 — Time Series Prediction

Dataset	Metric	LSTM	CT-RNN	Neural ODE	CT-GRU	LTC	SRNN (ours)	Verdict
Gesture	accuracy	64.57%	59.01%	46.97%	68.31%	69.55%	43.33%	[WARN] Below all [GOOD] Beats all [OK] Com- petitive [EARLY] Too early [EARLY] Approach- ing LSTM [EARLY] Suspi- ciously good [WARN] Below all
Occupancy	accuracy	93.18%	94.54%	90.15%	91.44%	94.63%	98.71%	
Activity (HAR)	accuracy	95.85%	95.73%	97.26%	96.16%	95.67%	94.55%	
	accuracy	98.41%	96.73%	97.61%	98.27%	97.57%	41.23%*	
Seq. MNIST	accuracy							
Traffic	MSE	0.169	0.224	1.512	0.389	0.099	0.166*	[EARLY] Approach- ing LSTM [EARLY] Suspi- ciously good [WARN] Below all
Power	MSE	0.628	0.742	1.254	0.586	0.642	0.026*	
Ozone	F1-score	0.284	0.236	0.168	0.260	0.302	0.129	

Hasani reports mean ± std over n=5 seeds with full training (50–200 epochs). Our numbers are single seed, 20 epochs.
 * Still running — current best, not final.

Table 4 — Person Activity (1st Setting)

Algorithm	Accuracy
LSTM	83.59% ± 0.40
CT-RNN	81.54% ± 0.33
CT-GRU	85.27% ± 0.39
LTC	85.48% ± 0.40

Algorithm	Accuracy
SRNN (ours)	[CRASHED] OOM crash

Table 6 — Half-Cheetah Dynamics

Algorithm	MSE
LSTM	2.500 ± 0.140
CT-RNN	2.838 ± 0.112
CT-GRU	3.014 ± 0.134
LTC	2.308 ± 0.015
SRNN (ours)	[FAILED] Not launched

Epoch-by-Epoch Training Curves

HAR — Activity Recognition [DONE] (Best @ epoch 19: valid 98.77%, test 94.55%)

Epoch	Train Loss	Train Acc	Valid Acc	Test Acc	
0	1.43	50.23%	13.78%	15.53%	
1	0.82	84.51%	63.57%	60.22%	
2	0.42	90.81%	92.09%	89.37%	
3	0.25	94.21%	94.95%	90.19%	
4	0.20	95.34%	93.86%	85.83%	
5	0.19	94.90%	96.04%	93.46%	
6	0.20	95.65%	96.04%	93.73%	
7	0.17	96.49%	97.00%	90.19%	
8	0.18	96.80%	96.86%	93.19%	
9	0.17	96.16%	93.86%	88.83%	
10	0.24	91.85%	95.63%	85.83%	<- dip (LR taper region)
11	0.26	89.49%	90.18%	86.65%	
12	0.20	93.01%	81.31%	85.83%	
13	0.13	97.12%	98.50%	95.10%	<- recovery
14	0.15	96.23%	98.36%	94.82%	
15	0.13	96.46%	98.36%	93.46%	
16	0.10	98.13%	95.23%	93.73%	
17	0.09	98.21%	98.50%	94.55%	
18	0.09	98.28%	97.68%	94.82%	
19	0.08	98.41%	98.77%	94.55%	<- BEST

Gesture [DONE] (Best @ epoch 8: valid 55.00%, test 43.33%)

Epoch	Train Loss	Train Acc	Valid Acc	Test Acc	
0	1.79	18.23%	25.00%	17.78%	
1	1.60	30.73%	25.00%	17.78%	
2	1.45	46.35%	40.00%	27.78%	
3	1.44	44.53%	43.33%	40.00%	
4	1.32	50.00%	43.33%	43.33%	
5	1.27	49.22%	46.67%	43.33%	
6	1.25	49.48%	48.33%	42.22%	
7	1.23	53.39%	51.67%	42.22%	
8	1.20	52.08%	55.00%	43.33%	<- BEST
9	1.15	55.47%	53.33%	46.67%	
...	(plateaus through epoch 19)				
19	1.02	61.20%	48.33%	45.56%	

Occupancy [DONE] (Best @ epoch 2: valid 98.89%, test 98.71%)

Epoch	Train Loss	Train Acc	Valid Acc	Test Acc
0	0.45	79.62%	69.33%	75.60%

1	0.12	96.93%	82.51%	72.82%	
2	0.05	98.74%	98.89%	98.71%	<- BEST
3	0.05	98.71%	98.77%	94.25%	
4	0.04	98.73%	98.89%	96.00%	
...	(overfits: valid stable ~98.7%, test degrades)				
19	0.02	99.29%	98.77%	89.48%	

Ozone [DONE] (Best @ epoch 9: valid F1 0.121, test F1 0.129)

Epoch	Train Loss	Valid F1	Prec	Recall	Test F1	
0	0.22	0.0142	6.25%	0.80%	0.0164	
1	0.15	0.0138	5.00%	0.80%	0.0161	
2	0.14	0.1185	6.30%	100.0%	0.1259	
3	0.14	0.1185	6.30%	100.0%	0.1259	
...	(plateaus – predicts positive class indiscriminately)					
9	0.13	0.1207	6.43%	97.60%	0.1293	<- BEST
19	0.13	0.1198	6.38%	97.60%	0.1298	

Traffic [RUNNING] (Best so far @ epoch 10: valid MSE 0.166, test MSE 0.168)

Epoch	Train Loss	Valid MSE	Valid MAE	Test MSE	Test MAE	
0	1.2130	1.9826	1.1386	2.0123	1.1486	
1	0.7454	0.9832	0.8610	0.9933	0.8662	
2	0.5746	0.3748	0.4840	0.3801	0.4872	
3	0.2653	0.2950	0.4133	0.3042	0.4201	
4	0.1907	0.2115	0.3548	0.2163	0.3585	
5	0.1816	0.1814	0.3196	0.1845	0.3231	
6	0.1782	0.1729	0.3095	0.1768	0.3129	
7	0.2780	0.2032	0.3395	0.2079	0.3422	<- spike
8	0.2680	0.4344	0.5512	0.4337	0.5517	<- spike
9	0.1844	0.2196	0.3634	0.2211	0.3651	
10	0.1830	0.1662	0.3061	0.1684	0.3080	<- best
11	0.1642	0.1896	0.3124	0.1921	0.3160	

SMnist [RUNNING] (2 epochs completed)

Epoch	Train Loss	Train Acc	Valid Acc	Test Acc
0	2.06	25.36%	10.45%	10.10%
1	1.20	58.24%	42.05%	41.23%

Power [RUNNING] (2 epochs completed)

Epoch	Train Loss	Valid MSE	Valid MAE	Test MSE	Test MAE
0	0.2045	1.4023	0.7983	1.3874	0.7966
1	0.0043	0.0263	0.0783	0.0259	0.0785

Analysis & Issues

[GOOD] Strong Results

- Occupancy (98.71% test accuracy):** Beats all models in Hasani Table 3, including LTC (94.63%). Converged extremely fast (best at epoch 2). Shows overfitting after: valid stays ~98.8% but test degrades from 98.71% -> 89.48% by epoch 19. Early stopping is critical for this task.
- HAR (94.55% test accuracy):** Competitive with Hasani’s LTC (95.67%). Valid accuracy reached 98.77% — the gap to test (94.55%) suggests some overfitting. With more seeds and full epochs, could match or exceed.
- Power (MSE 0.026 at epoch 1):** 22× better than Hasani’s best (0.586 CT-GRU). We verified both scripts use identical z-score normalization and per-timestep MSE — the metrics should be directly comparable. Needs more epochs to confirm this isn’t transient.
- Traffic (MSE 0.166 at epoch 10):** Already matching LSTM (0.169) and approaching LTC (0.099). Still improving with 8 epochs remaining.

[WARN] Needs Investigation

5. **Gesture (43.33% test accuracy):** Below all Hasani models including Neural ODE (46.97%). Learning curve shows continued improvement through epoch 19 (train acc 61.2%) but valid/test plateau. Possible causes: seq_len=32 may be too short for this dataset with SRNN dynamics, or 20 epochs insufficient. Note: Hasani trained 200 epochs.
6. **Ozone (F1 0.129):** Below all Hasani models (worst was Neural ODE at 0.168). The model achieves ~97.6% recall but only ~6.4% precision — it's predicting nearly everything as positive. This is a heavily imbalanced binary classification task. Likely needs: class weighting adjustments, threshold tuning, or different training regime.

[CRASHED] Failed / Not Run

7. **Person (OOM):** Julia process killed by Linux OOM-killer (exit code 137) on n4d-highmem-2 (16 GB) within 51 minutes, before completing first epoch. RSS exceeded available RAM. Fix: Updated person.env to use n4d-highmem-4 (32 GB, 4 vCPU).
8. **Cheetah (not launched):** 9th experiment couldn't launch due to external IPv4 address quota of 8. Fix: Migrate to IAP tunneling with --no-address VMs (planned).

Timing & Cost

Experiment	Per-Epoch Time	20-Epoch Time	Est. 50-Epoch Time
HAR	~15 min	305 min	~12.7 hr
Gesture	~2 min	38 min	~1.6 hr
Occupancy	~17 min	335 min	~14.0 hr
Ozone	~2 min	43 min	~1.8 hr
Traffic	~50 min	— (running)	~41.7 hr
SMnist	~3.5 hr	— (running)	~7.3 days
Power	~4.5 hr	— (running)	~9.4 days
Person	—	OOM	—
Cheetah	—	not run	—

SMnist and Power are extremely slow — 3.5 and 4.5 hours per epoch respectively. At 50 epochs, they'd run for 7–9 days each on a single VM. This needs investigation: likely caused by very long sequences (SMnist: 784 steps) or large datasets (Power: ~2M samples).

Next Steps

1. **Fix IPv4 quota bottleneck:** Switch to IAP tunneling + Cloud NAT (VMs with --no-address)
2. **Relaunch Person** on n4d-highmem-4 (env already updated)
3. **Launch Cheetah** (was blocked by IP quota)
4. **Investigate slow experiments:** SMnist and Power epoch times need optimization
5. **Investigate Gesture and Ozone** underperformance
6. **Full production run:** 5 seeds × 9 experiments = 45 VMs (needs IAP for >8 concurrent)
7. **Add final_metrics.json** output to training scripts (currently only training_log.txt)